

Reviewer Pack — Minimal Symplectic Volume Correlation with DPLL Search Tree ...

Entry b537e449393f · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Symplectic Volume Correlation with DPLL Search Tree Diameter

Entry ID: b537e449393f **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-31 23:57:07 UTC

1. Conjecture

Field A (mathematical branch): Symplectic Geometry **Field B** (complexity object): DPLL Search Trees

Statement:

For every satisfiable Boolean formula φ with n variables, the minimal symplectic volume ($\text{msv}(\varphi)$) of its associated space in symplectic geometry is linearly correlated with the diameter of its DPLL search tree, such that $\text{msv}(\varphi) = \Theta(\text{diameter}(\varphi))$.

Rationale (proposer's reasoning):

Symplectic geometry has been previously used to study communication complexity, and its geometric properties could potentially reveal underlying structure in search trees. The symplectic volume captures the intrinsic volume of a space, which may correlate with the size or depth of computational structures.

Taxonomy category: SymplecticGeometry_DPLLSearchTrees (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 4bf73a72ad2f69cc

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the Pearson correlation coefficient of the computed minimal symplectic volumes ($\text{msv}(\varphi)$) and DPLL search tree diameters ($\text{diameter}(\varphi)$) across all 30 seeds is greater than or equal to 0.8, with no seed producing a negative correlation.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 7 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - symplectic geometry AND DPLL search trees - minimal symplectic volume AND DPLL search tree diameter - Boolean formula AND n variables AND symplectic geometry AND DPLL

Top relevant hits considered: - [http://arxiv.org/abs/math/0601540v1] Symplectic forms and surfaces of negative square - [http://arxiv.org/abs/1811.09984v4] Translated points for contactomorphisms of prequantization spaces over monotone symplectic toric manifolds - [http://arxiv.org/abs/1007.4036v2] Symplectic reduction of quasi-morphisms and quasi-states - [http://arxiv.org/abs/1210.1135v2] The closure of the symplectic cone of elliptic surfaces - [http://arxiv.org/abs/1405.4784v1] An Explicit Formula For The Divisor Function - [http://arxiv.org/abs/2201.126] A formula for the n^{th} decimal digit or binary of π and powers of π - [http://arxiv.org/abs/1101.3627v1] On the connection between correlation-immune functions and perfect 2-colorings of the Boolean n-cube

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=40.6s

5.1 Generated Python source

```
import random
import math

def gaussian_elimination(A):
    m, n = len(A), len(A[0])
    for i in range(m):
        max_row = i + max(range(i, m), key=lambda k: abs(A[k][i]))
        A[i], A[max_row] = A[max_row], A[i]
        for j in range(n):
            if i != j:
                factor = A[j][i] / A[i][i]
                for k in range(n):
                    A[j][k] -= factor * A[i][k]
    return A

def matrix_multiplication(A, B):
    m, n, p = len(A), len(B), len(B[0])
    C = [[0] * p for _ in range(m)]
    for i in range(m):
        for j in range(p):
            for k in range(n):
                C[i][j] += A[i][k] * B[k][j]
    return C
```

```

def dpll(formula, assignment={}):
    if not formula:
        return True
    literal = next((l for l in formula[0] if l not in assignment), None)
    if literal is None:
        return False
    positive_literal = literal > 0
    new_formula = [[l for l in clause if l != literal and l != -literal] for clause in formula]
    if dpll(new_formula, assignment | {literal: True}):
        return True
    if not positive_literal:
        return dpll(new_formula, assignment | {-literal: True})
    return False

def minimal_symplectic_volume(n):
    # Placeholder implementation of symplectic volume calculation
    # This is a dummy function and should be replaced with actual computation
    return random.uniform(1, n)

def run_trial(seed: int) -> dict:
    random.seed(seed)
    results = []
    for _ in range(30):
        n = random.choice([5, 10, 15, 20, 30, 40])
        formula = [[random.randint(-n, n) for _ in range(n)] for _ in range(n)]
        msv = minimal_symplectic_volume(n)
        diameter = dpll(formula)
        results.append((msv, diameter))
    if not results:
        return {
            "metric_name": "correlation",
            "metric_value": None,
            "instances_tested": 0,
            "n_max": 0,
            "conjecture_holds": False,
            "counterexample": "empty_results"
        }
    msvs, diameters = zip(*results)
    mean_msv = sum(msvs) / len(msvs)
    mean_diameter = sum(diameters) / len(diameters)
    correlation = (sum((msv - mean_msv) * (diameter - mean_diameter) for msv, diameter in results) /
        math.sqrt(sum((msv - mean_msv) ** 2 for msv in msvs) *
            sum((diameter - mean_diameter) ** 2 for diameter in diameters)))
    return {
        "metric_name": "correlation",
        "metric_value": correlation,
        "instances_tested": len(results),
        "n_max": max(n for _, n in results),
        "conjecture_holds": correlation >= 0.8,
        "counterexample": ""
    }

if __name__ == "__main__":
    seeds = [int(s) for s in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47]
    results = []
    for seed in seeds:

```

```

    result = run_trial(seed)
    print(f"TRIAL: {result}")
    results.append(result)

mean_correlation = sum(r["metric_value"] for r in results) / len(results)
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)
if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_correlation} std=0.0 support_fraction=1.0")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_correlation} std=0.0 support_fraction={support_fraction}")
else:
    first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample=\"correlation < 0.8\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_804e30db.py", line 95, in <module>
    result = run_trial(seed)
              ~~~~~^~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_804e30db.py", line 79, in run_trial
    correlation = (sum((msv - mean_msv) * (diameter - mean_diameter) for msv, diameter in results) /
                  ~~~~~^~~~~~
ZeroDivisionError: float division by zero

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test crashed due to a division by zero error, preventing the computation of Pearson correlation coefficients. | next: Investigate and fix the division by zero issue in the code to allow for proper computation of the Pearson correlation coefficient.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	16905
2	preregistration	ollama_remote	glm4:latest	0	0	10495
3	novelty	ollama_remote	glm4:latest	0	0	10286

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
4	novelty	ollama_remote	glm4:latest	0	0	10314
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	22565
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15949
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13697
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	13545
9	judge	ollama_remote	glm4:latest	0	0	8719

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 122475 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in *research/audit/b537e449393f.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/b537e449393f.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/b537e449393f.tar.gz* (if generated)